

图 4.1-2

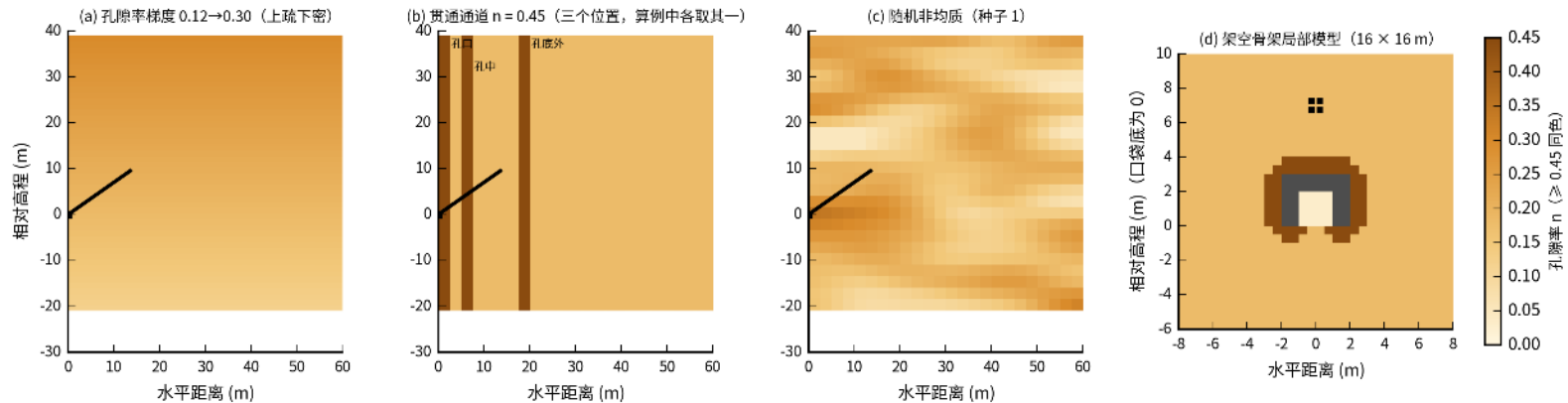


图 4.3-1

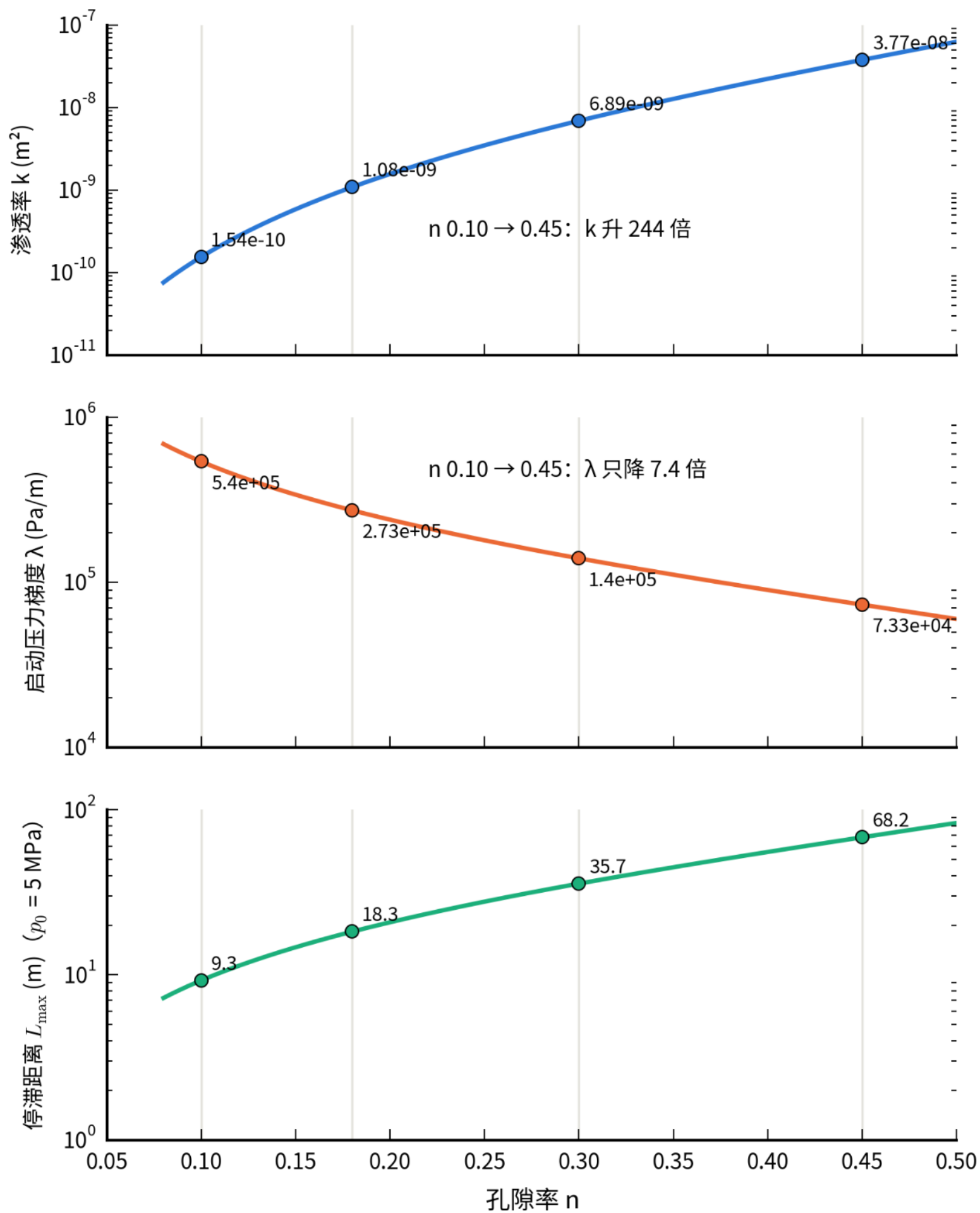


图 4.4-2

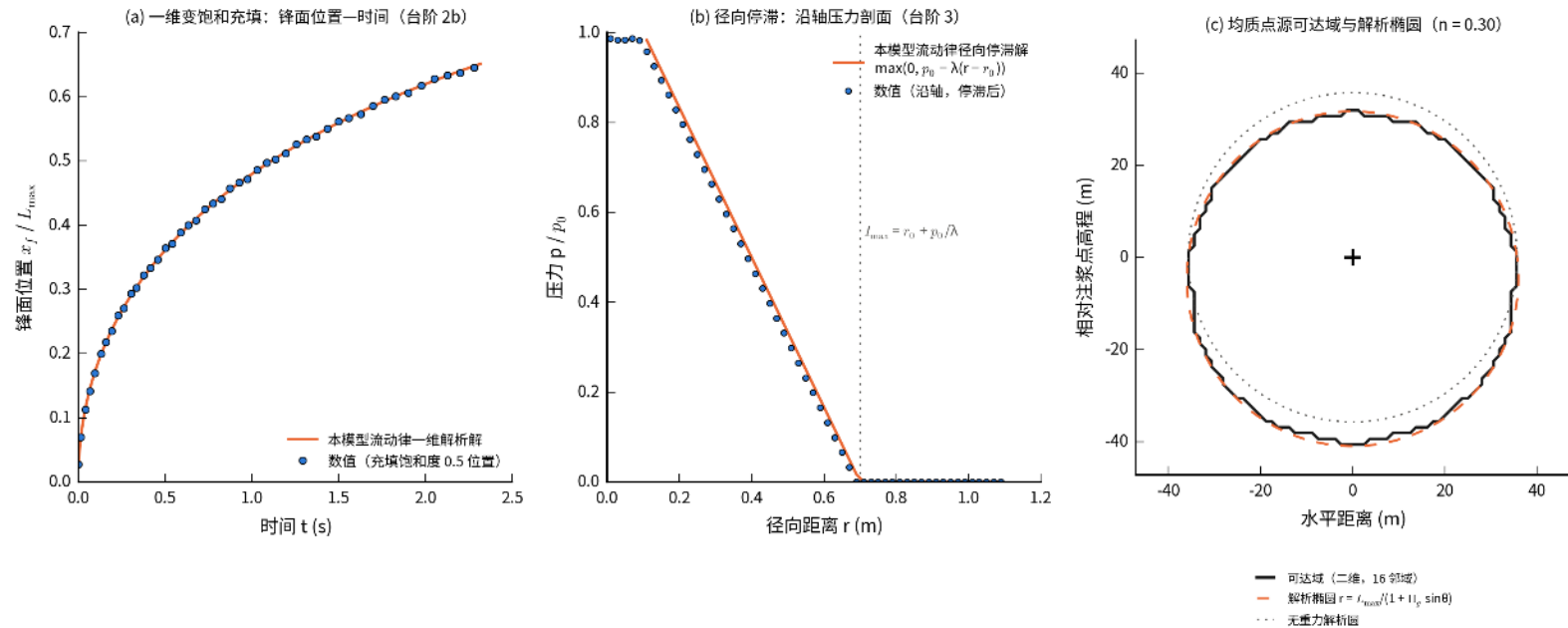
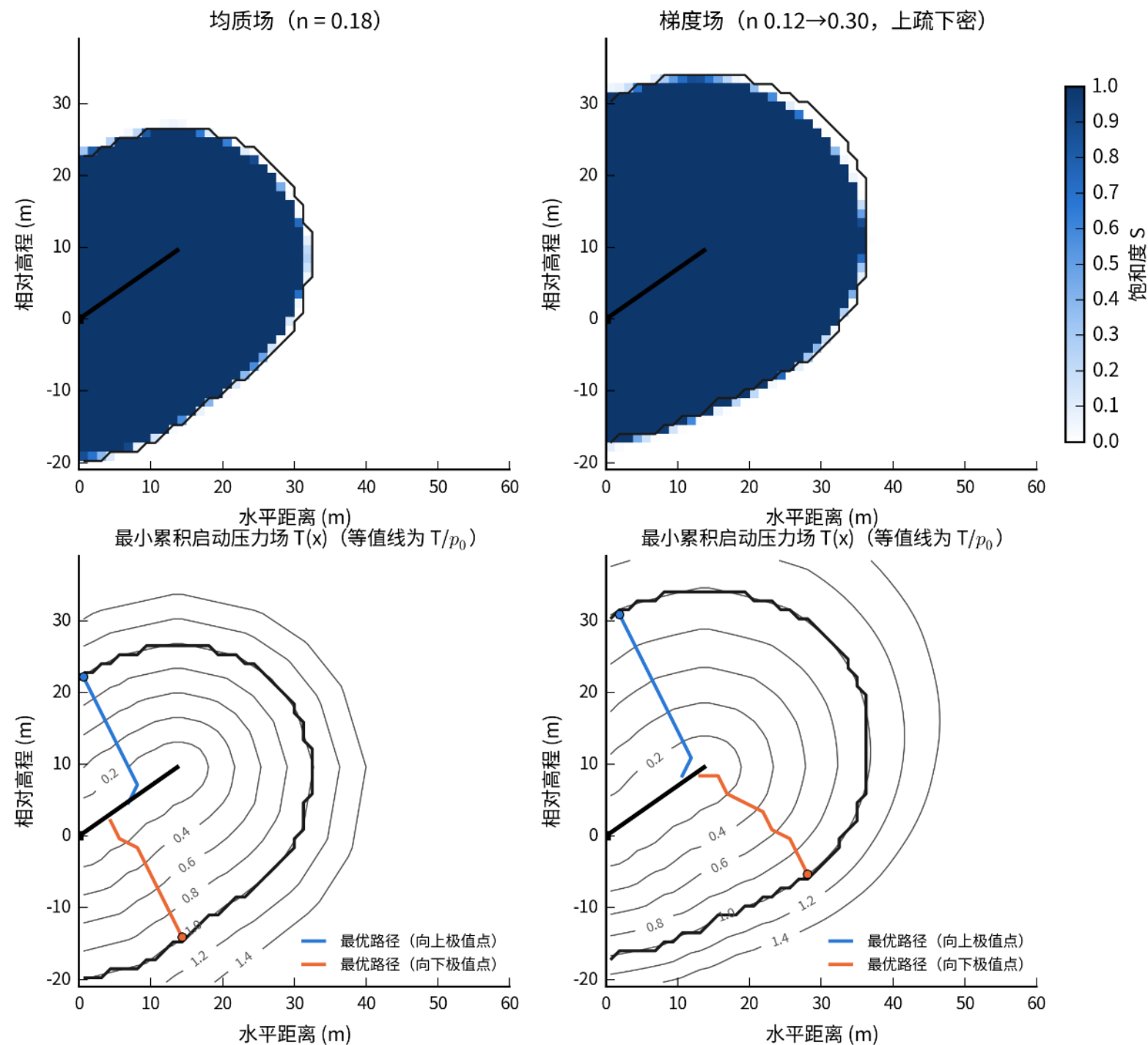


图 4.5-1



实线：可达域；色斑：充填域（停注时）；钻孔孔口位于左边界（坡面）

图 4.5-2

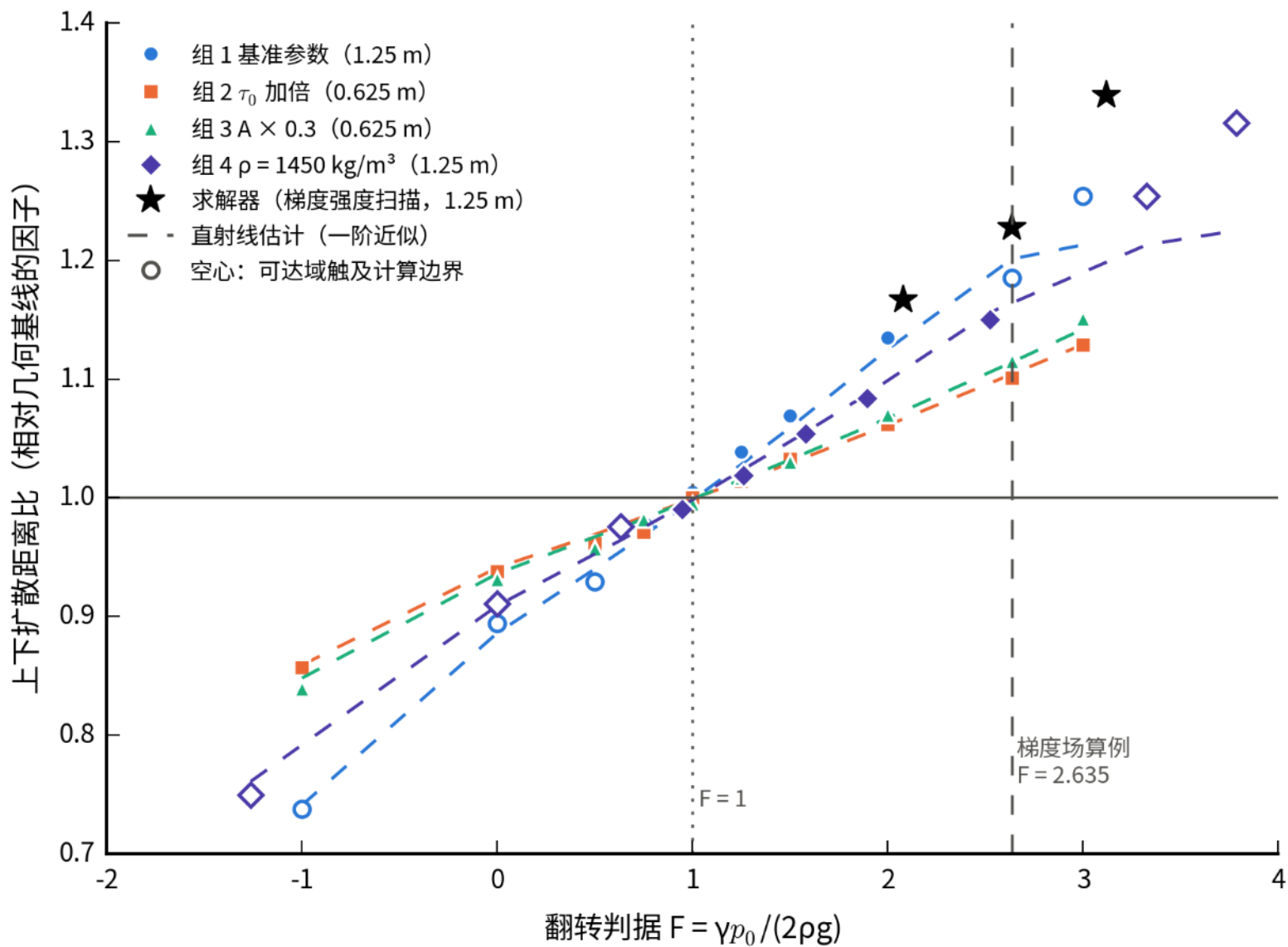


图 4.5-3

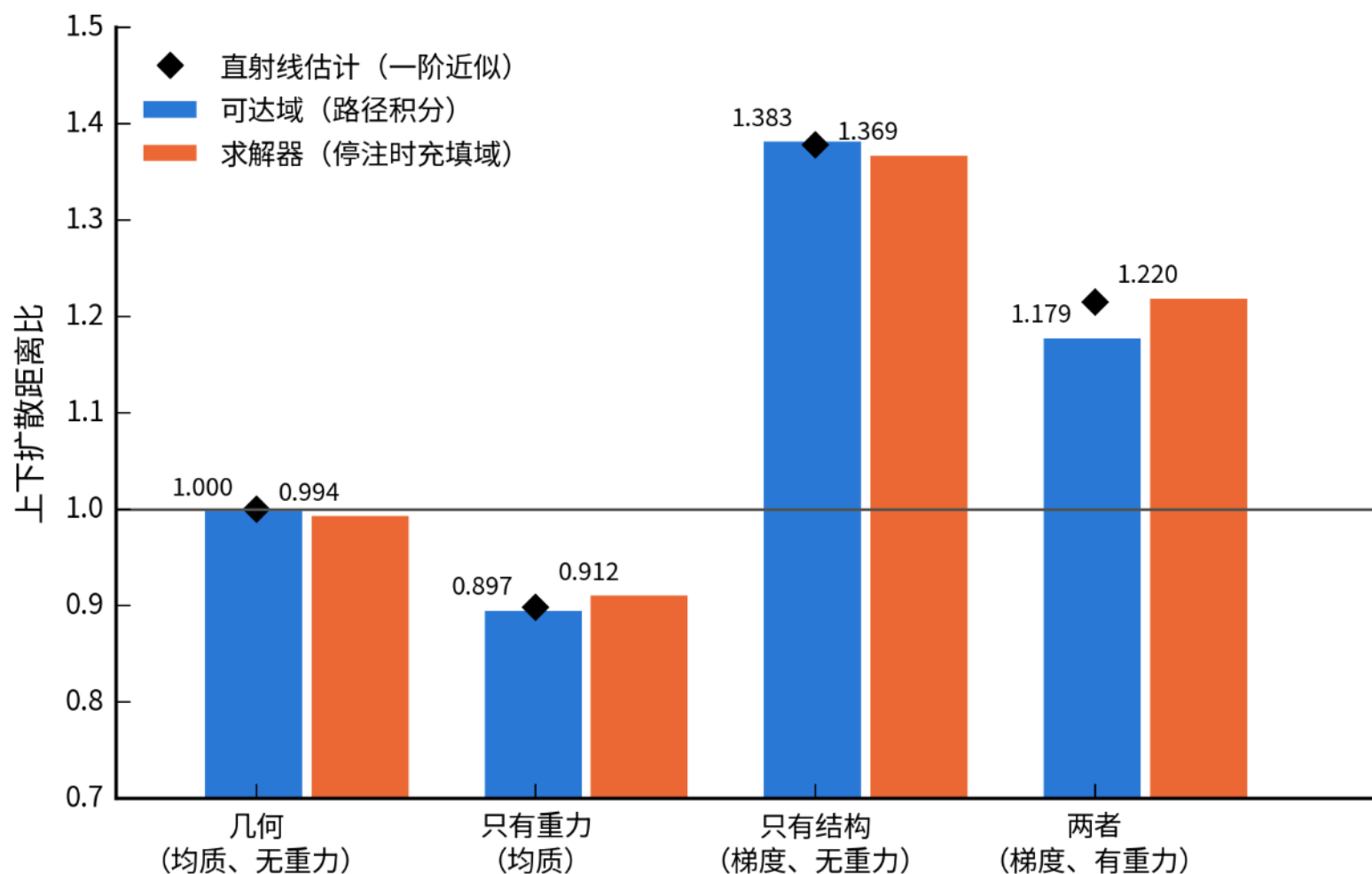


图 4.5-4

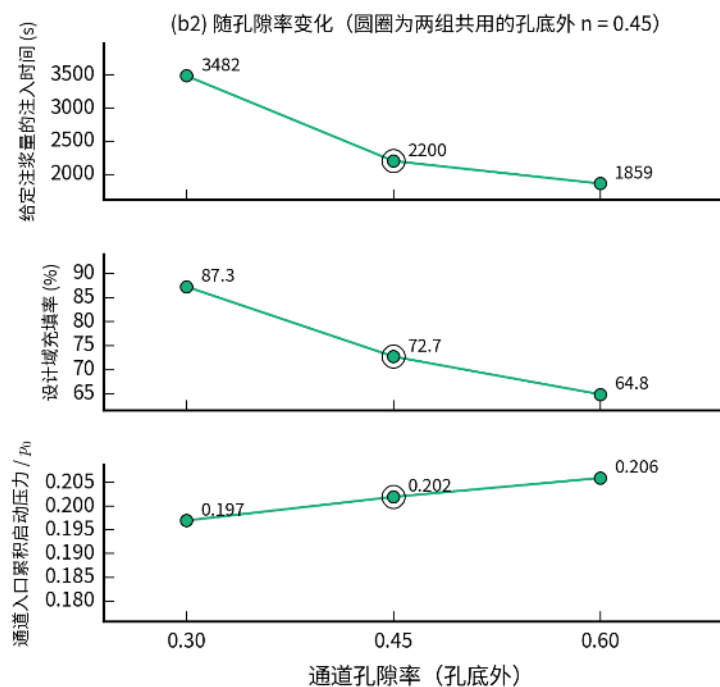
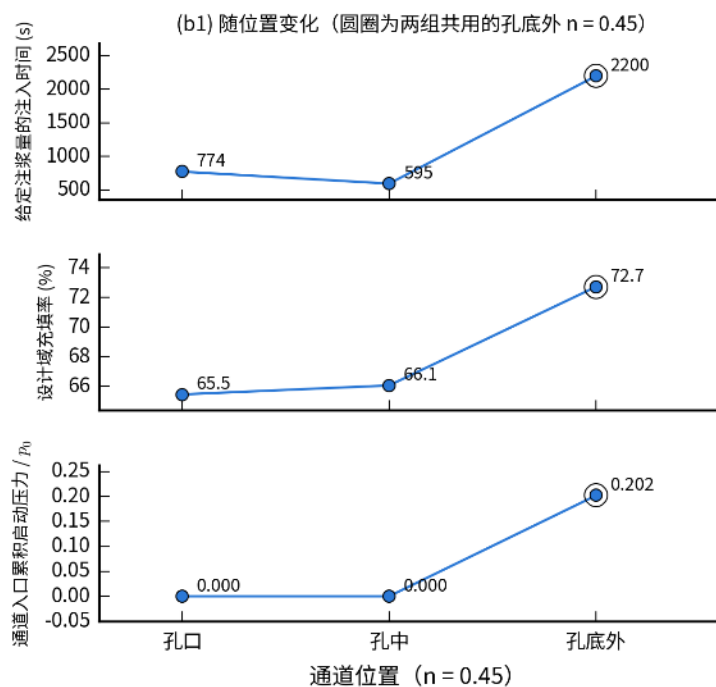
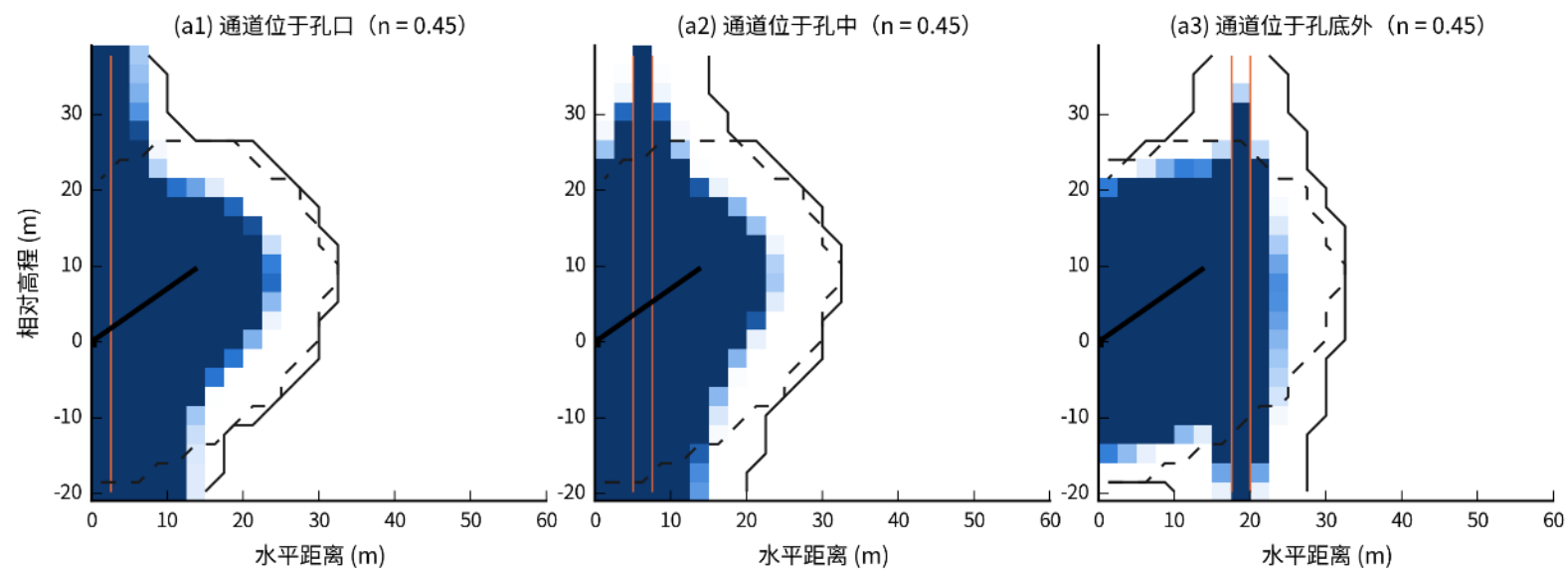
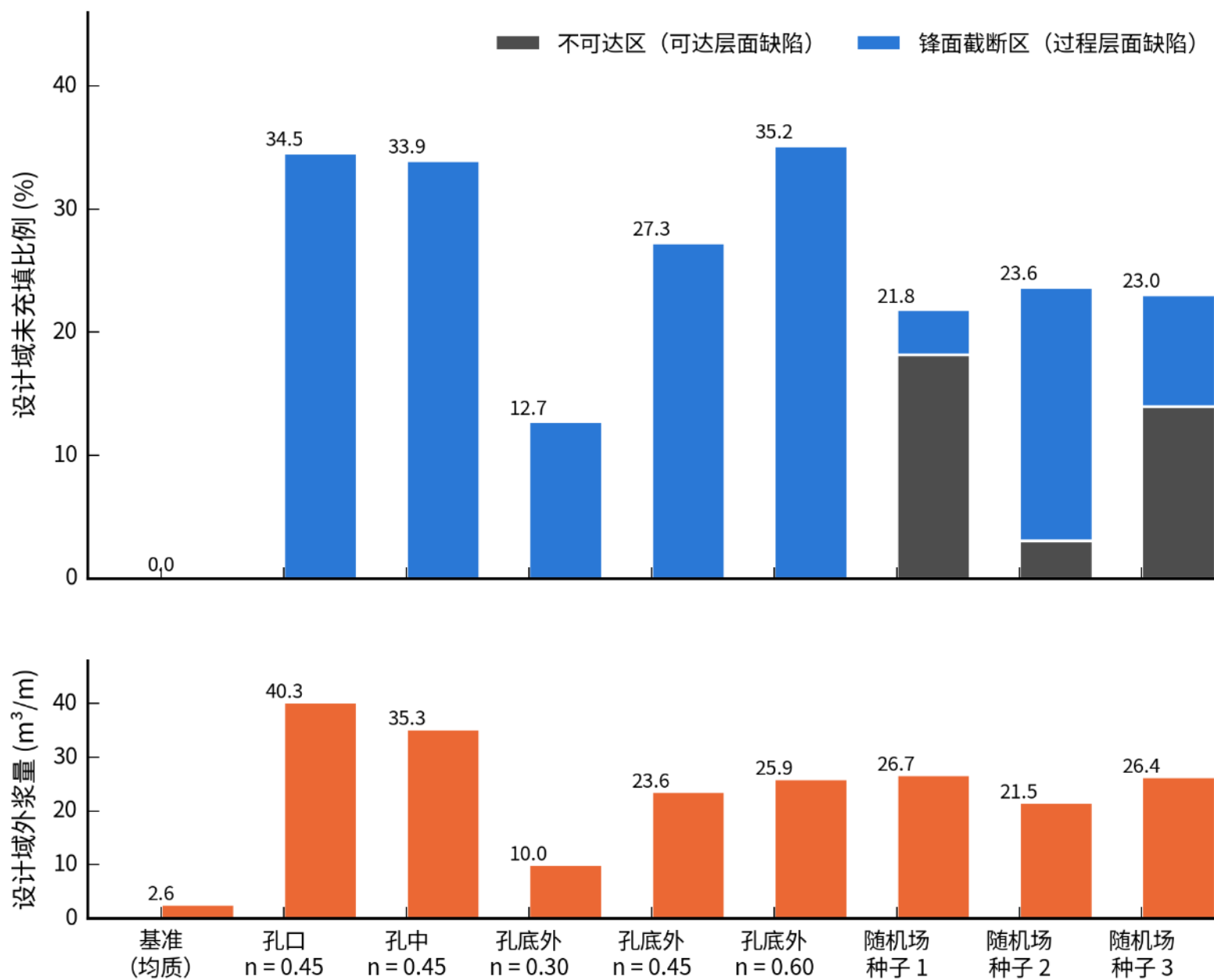
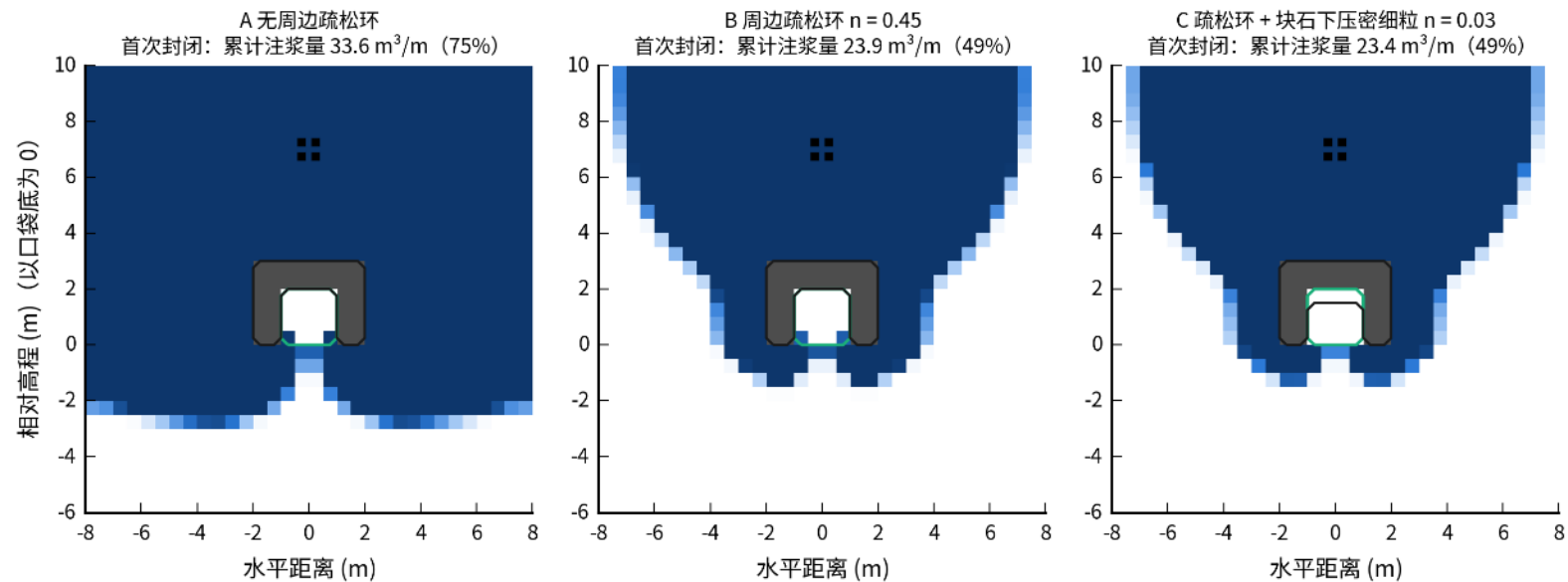


图 4.5-5



未充填比例的分母为设计域（不计钻孔单元，165 个单元）；设计域外浆量为设计域以外、计算域内的浆量（不计钻孔单元）

图 4.5-6



色斑: 充填域 (首次封闭时); 深灰: 块石; 绿线: 块石下口袋; 实线: 可达域; 黑方块: 注浆源

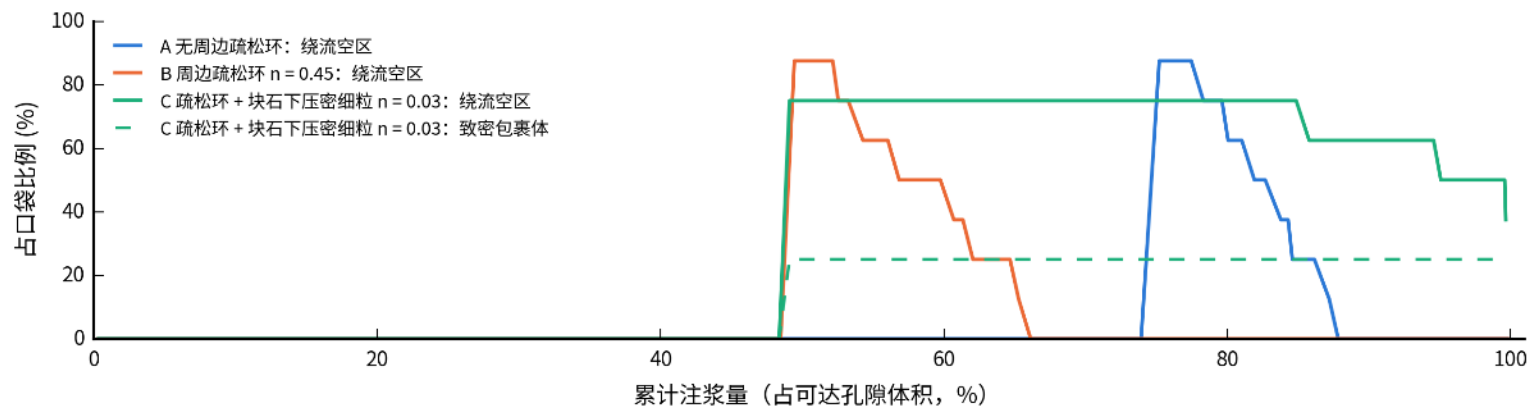


图 4.6-2

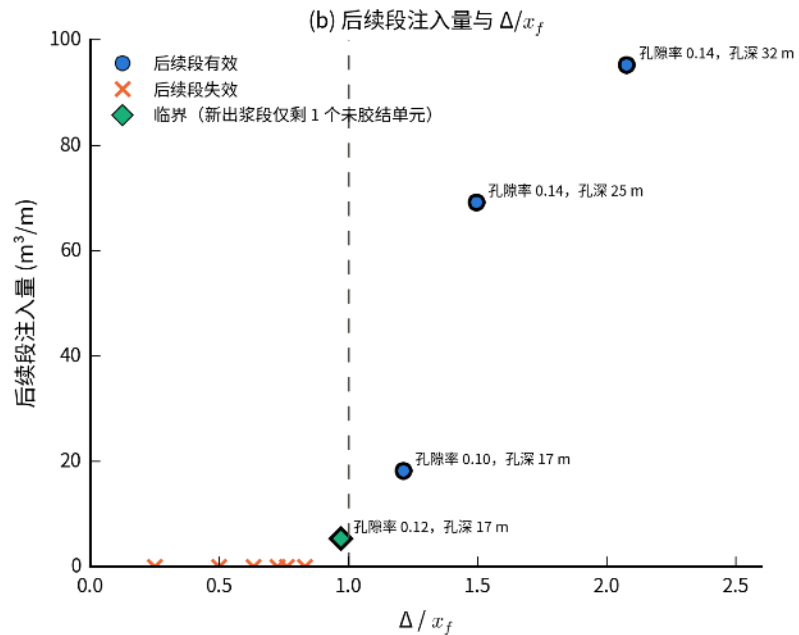
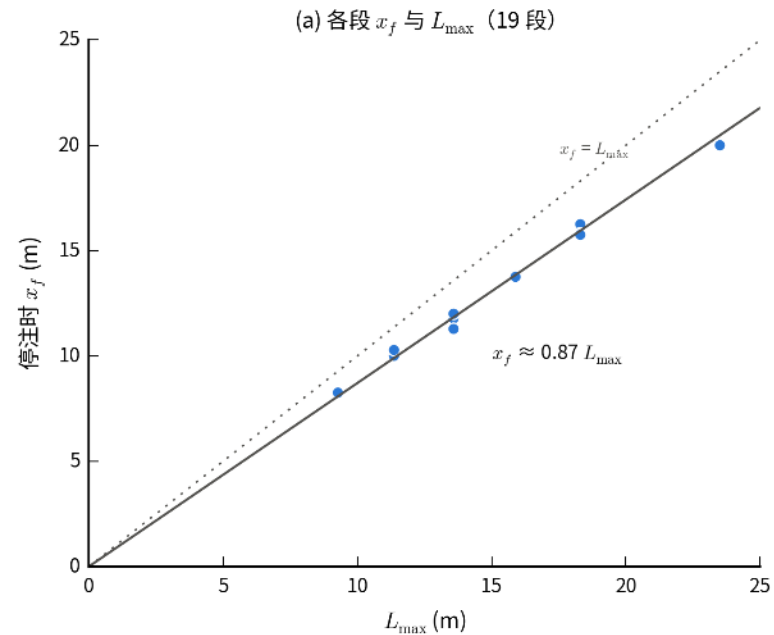


图 4.6-3

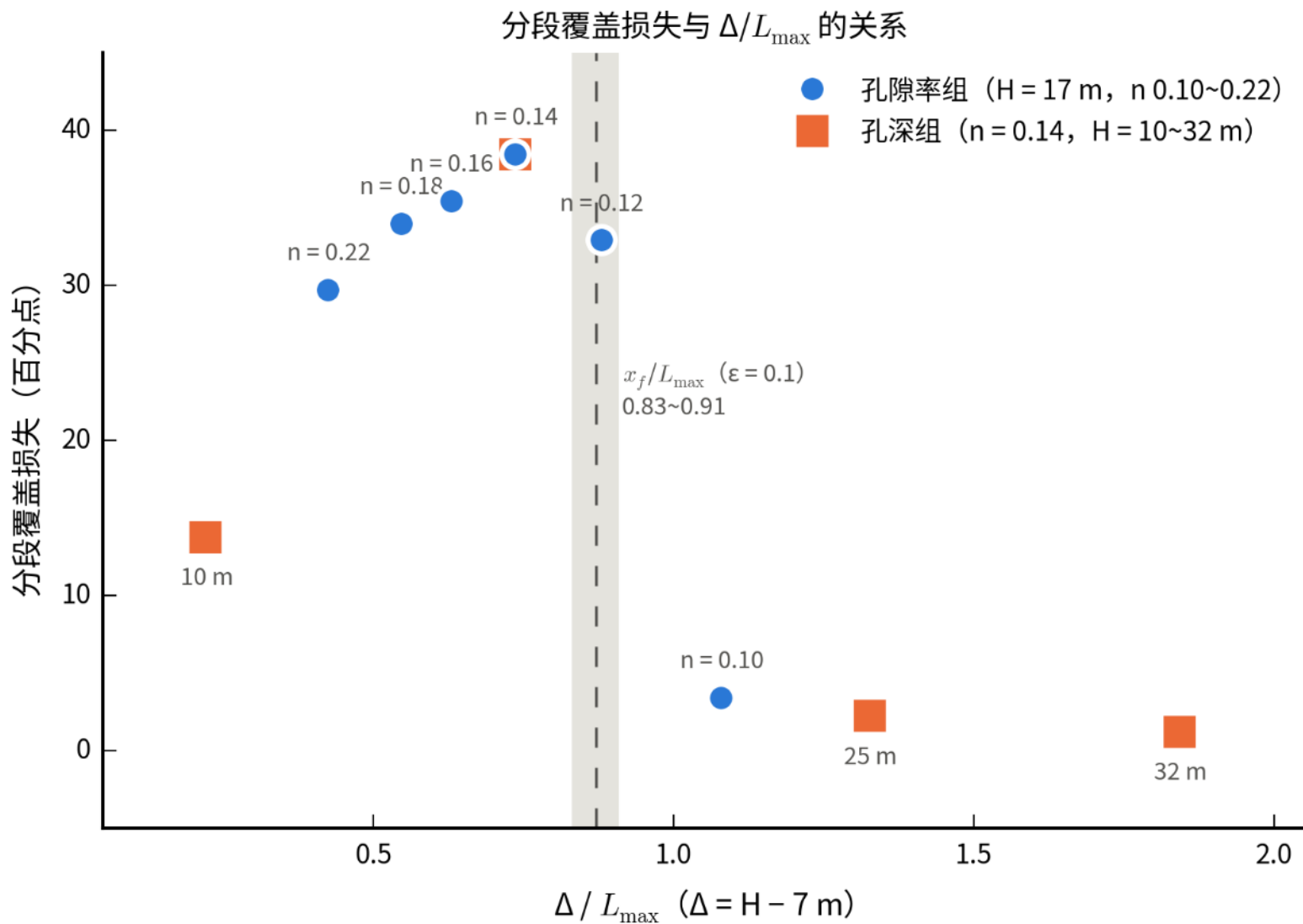
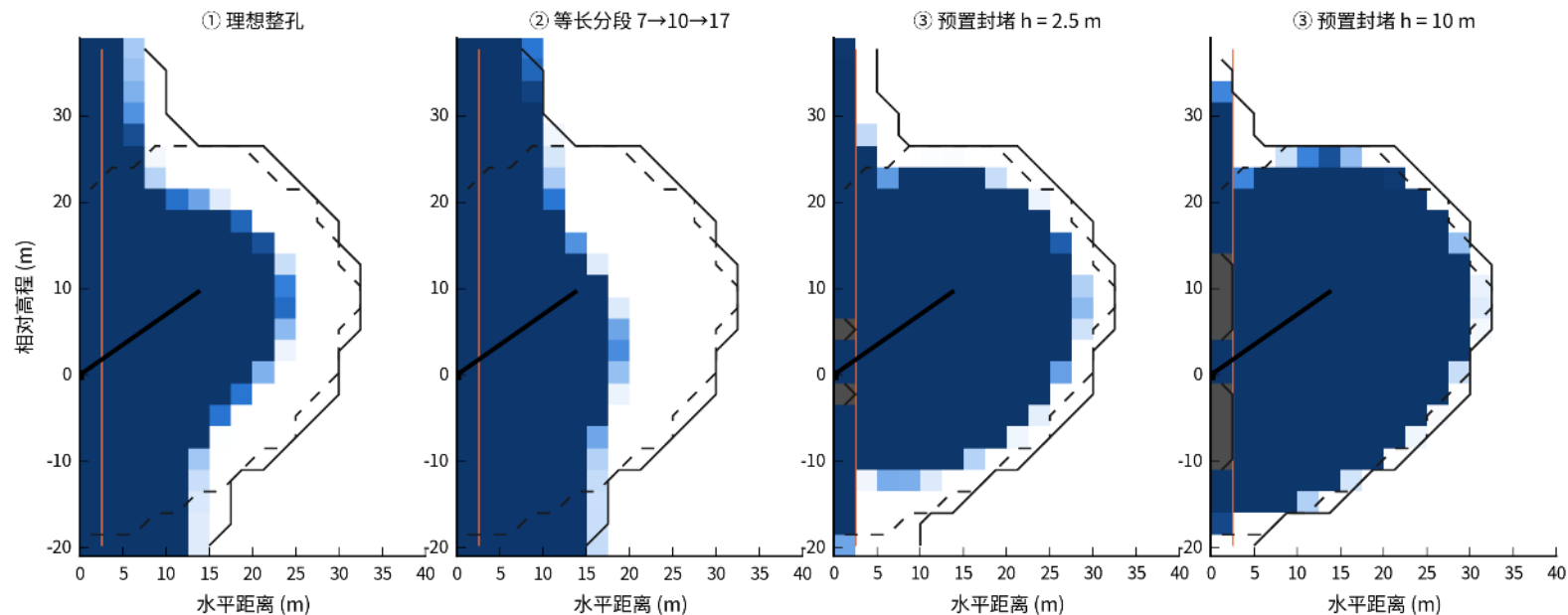


图 4.6-4



设计域充填率与设计域外浆量占比均不计钻孔单元；占比 = 设计域外浆量 ÷ 计算域内浆量（不计钻孔单元），不足注入量

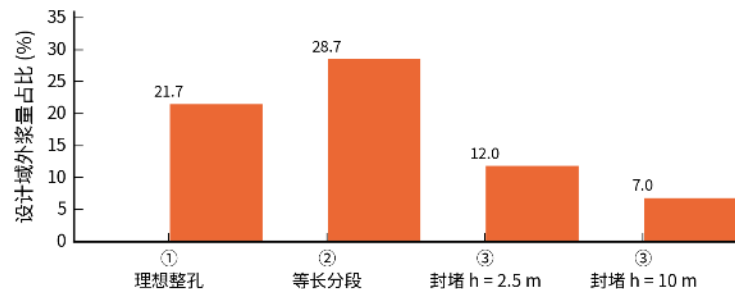
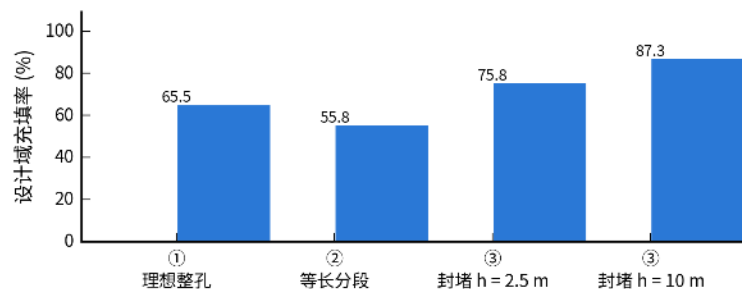
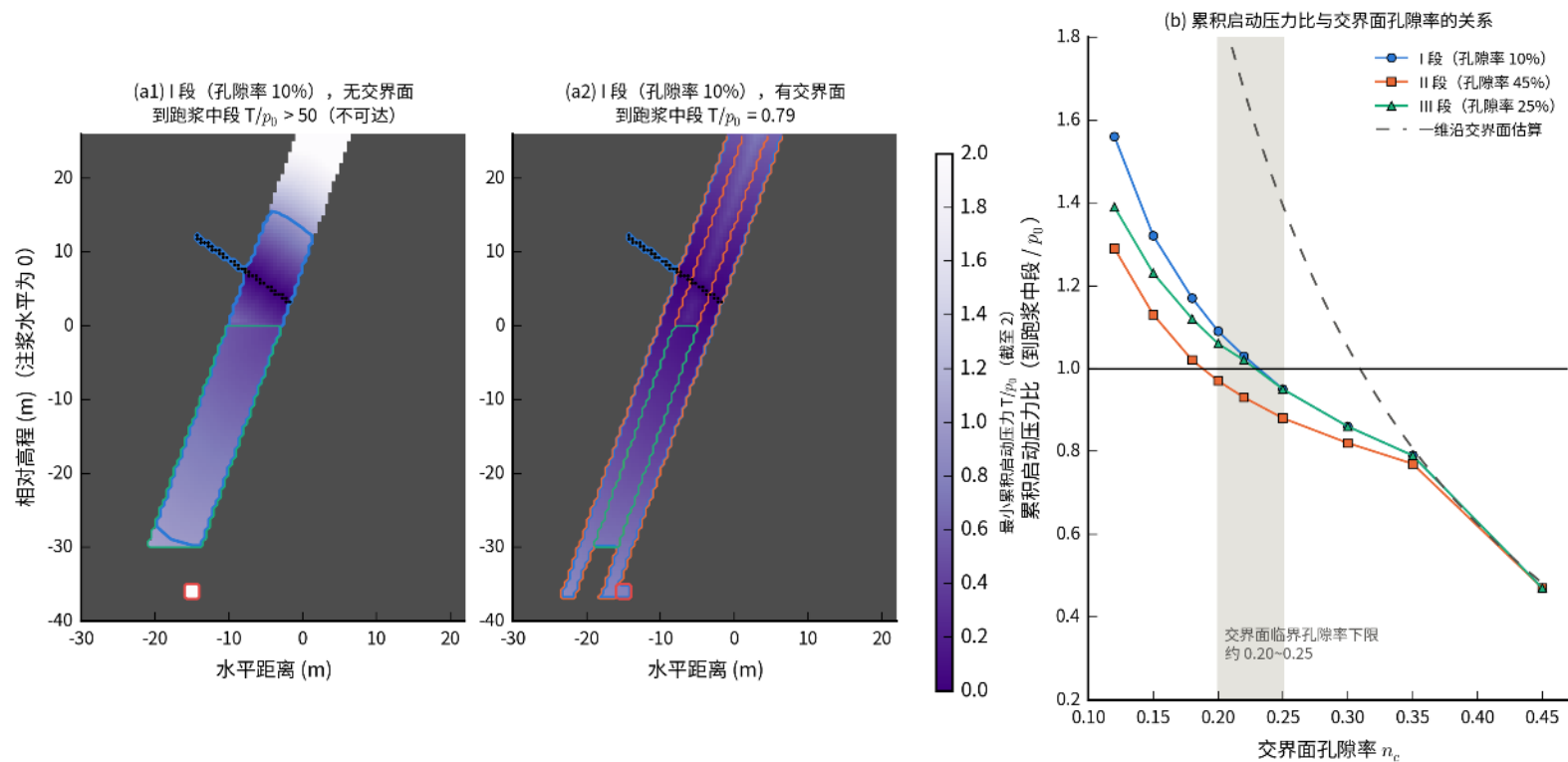
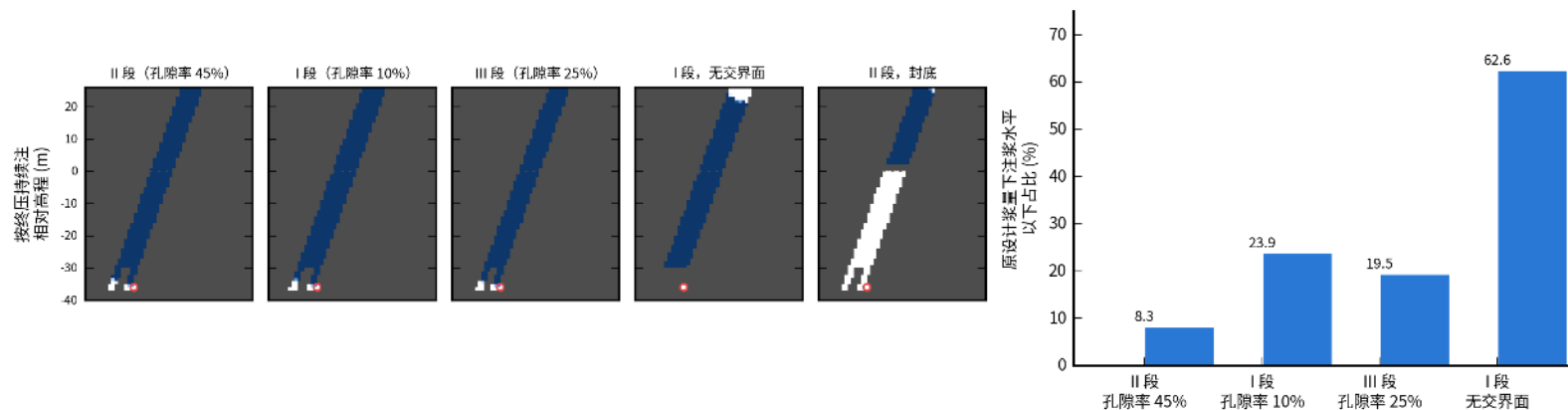
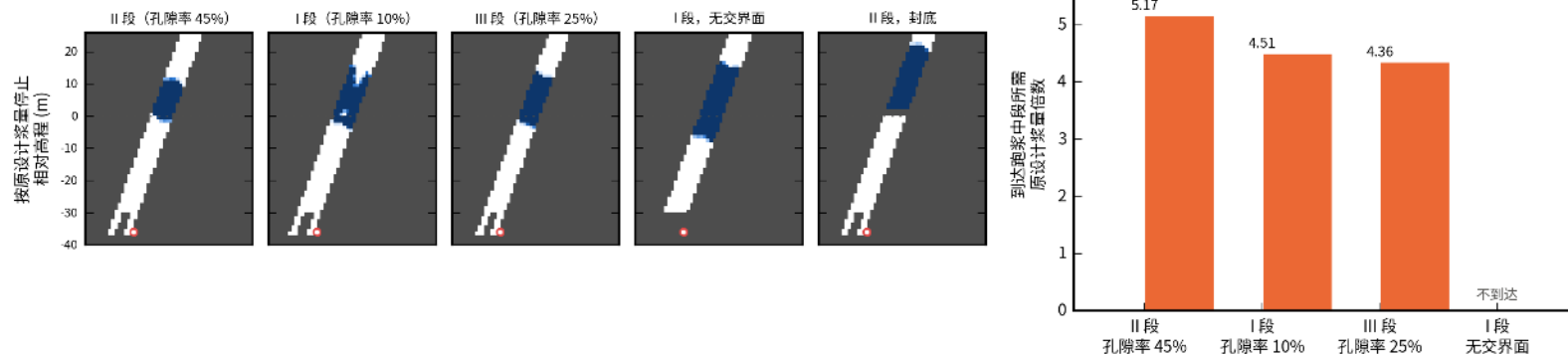


图 4.7-2



蓝线: 可达域边界 ($T = p_0$) ; 橙线: 交界面; 绿线: 存窿矿石; 红框: 跑浆中段; 黑点: 钻孔出浆段; 深灰: 围岩与未采矿体 (不透浆)

图 4.7-3



色斑: 充填域; 深灰: 围岩、未采矿体与封底; 红框: 跑浆中段; 点线: 注浆水平。上排 E 为封底后按变更设计浆量停止。